Sets (Part 1)

PreCalculus -I

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Applications of Sets

Typically application in computing involves

- Databases
- Data types in computer programming
- Constructing discrete structures
- Finite state machines
- Modeling computing machines
- Representing computational complexity of algorithms

What is a Set?

Definition: A set is an unordered collection of objects. The objects in a set are called the elements, or members, of the set. A set is said to *contain* its elements.

Examples:

- $\mathbb{Z}$: the set of integers
- Cities in Pakistan: $\{\text{Lahore}, \text{Karachi}, \text{Islamabad}, \ldots\}$
- Sets can contain non-related elements: $\{3, a, \text{red}, \text{Gilgit}\}$

Properties:

- Order does not matter: $\{1, 2, 3, 4, 5\} = \{3, 5, 2, 4, 1\}$
- Sets do not have duplicate elements
- It does not make sense to list the same element twice

Set Membership

- a is an element of the set A, denoted by $a \in A$.
- a is not an element of the set A, denoted by $a \notin A$.

Example:

- Let D be the set of students taking Discrete Mathematics.
- Ali is taking Discrete Mathematics: Ali $\in D$
- Saad is not taking Discrete Mathematics: Saad $\notin D$

Set Membership: More Examples

- $V = \{a, e, i, o, u\}$ $\quad a \in V, \quad b \notin V$
- $I = \{0, 1, 2, \ldots, 99\}$ $\quad 50 \in I, \quad 100 \notin I$
- $S = \{a, 2, \text{class}\}$ $\quad 2 \in S, \quad \text{room} \notin S$

Notation

1. Capital letters ($A, B, S, \ldots$) are used for sets.
2. Italic lower-case letters ($a, x, y, \ldots$) are used for elements.
3. The easiest way: list all the elements.

$$A = \{1, 2, 3, 4, 5\}$$

4. Not always feasible! We may use ellipsis ($\ldots$).

$$B = \{0, 1, 2, 3, \ldots\}$$

Possible confusion: $C = \{3, 5, 7, \ldots\}$ — what is next?

- If $C =$ odd integers greater than 2, next is 9.
- If $C =$ prime numbers greater than 2, next is 11.

Set Builder Notation

Set builder: characterise all elements in the set by stating the properties they must have.

Example:

$$O = \{x \mid x \text{ is an odd positive integer less than } 10\}$$

$$O = \{x \in \mathbb{Z}^+ \mid x \text{ is odd and } x < 10\}$$

$$O = \{1, 3, 5, 7, 9\}$$

Note

The vertical bar | means "such that".

Important Sets

Name	**Set**
Natural numbers	$\mathbb{N} = \{1, 2, 3, \ldots\}$
Integers	$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$
Positive integers	$\mathbb{Z}^+ = \{1, 2, 3, \ldots\}$
Rational numbers	$\mathbb{Q} = \{p/q \mid p, q \in \mathbb{Z},\ q \neq 0\}$
Real numbers	$\mathbb{R}$

Examples:

- $S_1 = \{\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}\}$ — S_1 has 4 elements, each of which is a set.
- $S_2 = \{x \mid x \in \mathbb{N} \text{ and } \exists k \in \mathbb{N},\ x = k^2\}$ — squares of natural numbers.

Equality of Sets

Let A and B be two sets.

Definition: A and B are equal if and only if they have the same elements, denoted $A = B$.

$$\boxed{A = B \iff \forall x\,(x \in A \leftrightarrow x \in B)}$$

Examples:

- $\{1, 2, 3\}$ and $\{3, 2, 1\} \quad \Rightarrow \quad \{1, 2, 3\} = \{3, 2, 1\}$
- $\mathbb{Z}^+$ and $\{0, 1, 2, \ldots\} \quad \Rightarrow \quad \mathbb{Z}^+ \neq \{0, 1, 2, \ldots\}$

The Universal Set

U is the universal set — the set of all elements (or the "universe") from which any given set is drawn.

Examples:

- For the set $\{-2,\ 0.4,\ 2\}$, U would be the real numbers.
- For the set $\{0, 1, 2\}$, U could be $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ depending on context.
- For the set of vowels of the alphabet, U would be all the letters of the alphabet.

Empty Set

Example:

$$S = \{x \mid x \in \mathbb{Z}^+ \text{ and } x < 0\} = \{\} = \emptyset$$

A set that has no elements is called the empty set, or null set.

Important distinction:

$$\boxed{\emptyset \neq \{\emptyset\}}$$

The left is the empty set; the right is a set containing one element (which happens to be the empty set).

Subset

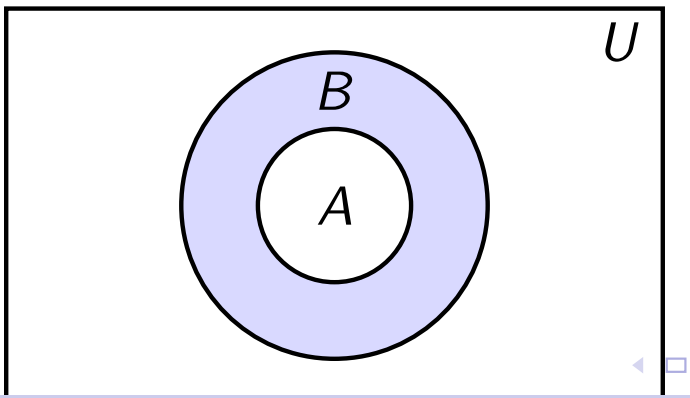
Let A and B be sets.

Definition: A is a subset of B if and only if every element of A is also an element of B, denoted $A \subseteq B$.

$$\boxed{A \subseteq B \iff \forall x\,(x \in A \to x \in B)}$$

Facts:

- For every set S: $\emptyset \subseteq S$
- For every set S: $S \subseteq S$



Subset and Equality

If

$$A \subseteq B, \quad \forall x\,(x \in A \rightarrow x \in B)$$

and

$$B \subseteq A, \quad \forall x\,(x \in B \rightarrow x \in A)$$

then

$$\boxed{A = B \iff \forall x\,(x \in A \leftrightarrow x \in B)}$$

Key idea: Two sets are equal if and only if each is a subset of the other.

Subset: Examples

- $\mathbb{Q}$ and $\mathbb{R}$: $\mathbb{Q} \subseteq \mathbb{R}$
- $\mathbb{N}$ and $\mathbb{Z}$: $\mathbb{N} \subseteq \mathbb{Z}$
- $A = \{x \mid x \in \mathbb{Z}^+,\ x < 10\} = \{1, 2, \ldots, 9\}$
- $B = \{x \mid x \in \mathbb{Z}^+,\ x \text{ is even},\ x < 10\} = \{2, 4, 6, 8\}$
- $B \subseteq A$

Proof: $\emptyset \subseteq S$ for Every Set S

Claim: For every set S, $\emptyset \subseteq S$.

Proof: We want to show

$$\forall x\,(x \in \emptyset \rightarrow x \in S).$$

$\emptyset$ contains no element, so $x \in \emptyset$ is **false**. The hypothesis of the conditional statement is false, so

$$x \in \emptyset \rightarrow x \in S$$

is **true** (vacuously).

Thus $\forall x\,(x \in \emptyset \rightarrow x \in S)$ is true.

$$\boxed{\emptyset \subseteq S}$$

Proof: $S \subseteq S$ for Every Set S

Claim: For every set S, $S \subseteq S$.

Proof: We want to show

$$\forall x\,(x \in S \rightarrow x \in S).$$

Two cases arise:

- If $x \in S$ is true, then hypothesis and conclusion are both true; $(x \in S \rightarrow x \in S)$ is true.
- If $x \in S$ is false, then hypothesis and conclusion are both false; $(x \in S \rightarrow x \in S)$ is true.

Thus $\forall x\,(x \in S \rightarrow x \in S)$ is true.

$$\boxed{S \subseteq S}$$

Proper Subset

Let A and B be sets.

Definition: A is a proper subset of B if and only if $A \subseteq B$ but $A \neq B$, denoted $A \subset B$.

$$\boxed{A \subset B \iff \forall x\,(x \in A \to x \in B) \wedge \exists x\,(x \in B \wedge x \notin A)}$$

Analogy: The difference between "subset" and "proper subset" is like the difference between "$\leq$" and "$<$" for numbers.

Proper Subset: Examples

Let

$$T = \{0, 1, 2, 3, 4, 5\}, \quad S = \{1, 2, 3\}.$$

S is a subset of T and $S \neq T$, so S is a proper subset of T.

Let $Q = \{4, 5, 6\}$. Then Q is neither a subset nor a proper subset of T (since $6 \notin T$).

Note: If S is a subset of T and $S \neq T$, then S is a proper subset of T.